

COMMENT

L.P. Fatti : Department of Statistics, University of the Witwatersrand, P.O. Wits, 2050.

I am concerned about the anomalous behaviour of the sample information $I_{\bar{X}}$, illustrated in Figures 3 and 6, when the data fall near the extreme of, or outside, the range permitted by the prior distribution.

This behaviour appears to be largely an artifact of the posterior probability of θ :

$$\pi(\theta|\bar{x}) = \ell(\bar{x}|\theta)\pi(\theta) / \int_{\theta} \ell(\bar{x}|\theta)\pi(\theta)d\theta$$

being zero outside the range permitted by its prior, despite the fact that $\bar{X}$ (the estimator of θ) falls there.

A further cause for concern is the dramatic (and seemingly arbitrary) effect which the nature of the prior distribution of θ (whether it is expressed in continuous or discrete form) has on the profile of I_X , as illustrated in Figures 6 and 8. In any practical situation, however, it is usually impossible to choose between a continuous prior and its discrete counterpart, and it is therefore disturbing that this seemingly minor variation in the prior assumptions has such a large effect.

In view of the above, it would seem that the usefulness of the sample information I_X as a practical measure of the amount of information contained in the data X , is rather limited.